

Aditya Kulkarni

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| Raleigh, NC (Open to Relocation)

PROFESSIONAL SUMMARY

MS CS candidate (NC State, May 2026) with production full-stack experience in TypeScript, React, Java, and REST APIs. At Isomer AI, shipped 10+ features across 7 production deployments -- 394 commits, agile sprint delivery, CI/CD with Docker and GitHub Actions, and Sentry incident response. Built transactional Java APIs (Spring Boot + PostgreSQL) with Testcontainers TDD, and real-time event-driven platforms with MongoDB, Redis, and Server-Sent Events. Proficient in TypeScript/JavaScript, React, Angular, Java, REST APIs, relational and NoSQL databases, and cloud-native CI/CD pipelines.

CORE COMPETENCIES

TypeScript / JavaScript

React / Angular

Java / REST APIs

Agile / Scrum Delivery

CI/CD Pipelines

PostgreSQL / MySQL / MongoDB

TDD & Automated Testing

Docker / Cloud Infrastructure

WORK EXPERIENCE

Isomer AI May 2025 - Present
Product Engineer -- Chicago, IL (Remote)

- Shipped **10+ production features in agile sprints**: RBAC, real-time event pipelines, document processing (PDF, email via Microsoft Graph API, S3 file storage); TypeScript, Node.js, PostgreSQL, Redis -- 394 commits, 7 production deployments, PR reviews across the engineering team
- Maintained **CI/CD pipelines** with Docker and GitHub Actions; used Sentry for incident monitoring and postmortem analysis across 7 releases; collaborated with product and design on requirements and task delivery

NC State University Apr 2025 - Present
Graduate Research Assistant -- Raleigh, NC

- Designed a cloud-based web platform on Google Cloud Run serving 500GB genomic datasets to 100+ researchers; collaborated with cross-functional teams on ML architecture, improving model accuracy by 12%

Mphasis (HP Company) Jun 2024 - Aug 2024
Software Developer Intern -- Mumbai, India

- Built 5+ Python/web application workflows including health claim automation (OCR, NLP, ML); improved accuracy by 25%; automated deployments on AWS (EC2, S3) and GCP with Grafana monitoring

Capgemini Jun 2023 - Aug 2023
AI/ML Intern -- Mumbai, India

- Led end-to-end SDLC for a distributed data pipeline using Hadoop + Spark and Kafka streaming; reduced latency by 20% and processing time by 15%

PROJECTS

Inventory Reservation Service Java + TDD

Transactional REST API in Java 21 + Spring Boot with PostgreSQL row-level locking preventing overselling. Idempotent endpoints, RFC 9457 error responses, scheduled cleanup jobs. TDD with Testcontainers -- concurrent reservation tests prove exactly one success and one 409 Conflict. Flyway migrations, Spring Security (RBAC), multi-stage Docker build, GitHub Actions CI.
Java 21, Spring Boot 3.5, PostgreSQL, Spring Security, Testcontainers, Docker, GitHub Actions

PulseRank -- Real-Time Leaderboard & Activity Feed Full Stack

Full-stack event-driven platform: NestJS REST API backend, Next.js React frontend, MySQL transactional writes (idempotency key, outbox pattern), MongoDB feed projections (TTL expiry), Redis sorted sets for sub-millisecond leaderboard reads. Live updates via Server-Sent Events; OpenTelemetry + Prometheus observability; GitHub Actions CI/CD.
TypeScript, NestJS, Next.js, React, MySQL, MongoDB, Redis, Docker, GitHub Actions

CryptoKnight -- Full-Stack Next.js Application React / Next.js

Full-stack Next.js 16 application with TypeScript, viem, and wagmi for a trustless AI trading agent. ERC-4337 Account Abstraction with Session Keys -- enforces on-chain policies (trade limits, expiration, instant revocation). Built end-to-end in 48 hours at HackNC State 2025.
Next.js 16, TypeScript, React, viem, wagmi, Python, Alchemy Account Kit

AlgoPulse -- Event-Driven Platform Distributed

Re-architected API-polling bottleneck to fault-tolerant event-driven model: Temporal.io durable workflows + Kafka message brokering. Scales to 2,000 async requests/second without state loss under failure scenarios.

TypeScript, Node.js, React, Temporal.io, Kafka, Supabase Edge Functions

EDUCATION

North Carolina State University -- MS Computer Science Aug 2024 - May 2026

Courses: Algorithms, Software Engineering, Object-Oriented Development, Operating Systems, Artificial Intelligence, Data Science

Dwarkadas J. Sanghvi College of Engineering -- BTech Computer Science Aug 2020 - May 2024

CGPA: 9.32/10 · Courses: DS & Algorithms, OS, Software Engineering, Machine Learning

SKILLS

Languages: TypeScript, JavaScript, Java, Python, SQL, C, R

Frontend & Full Stack: React, Next.js, Angular, NestJS, Node.js, REST APIs, Server-Sent Events, Tailwind CSS

Backend & Frameworks: Spring Boot, FastAPI, Temporal.io, Kafka, Spark

Databases: PostgreSQL, MySQL, MongoDB, Redis

Cloud & DevOps: AWS (EC2, S3), GCP (Cloud Run), Docker, Kubernetes, GitHub Actions, Terraform, OpenTelemetry, Sentry

Testing: TDD, Testcontainers, Jest, pytest, k6